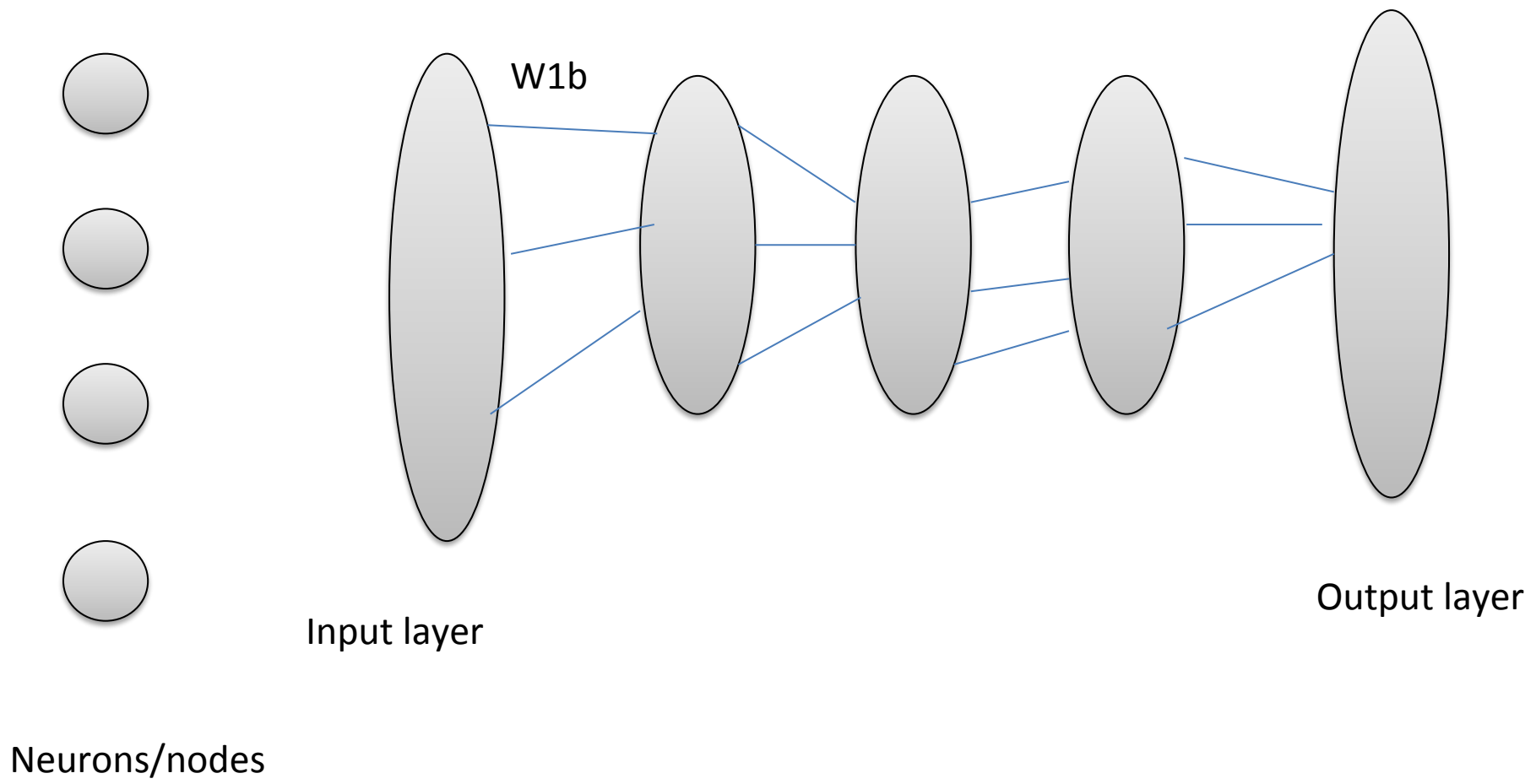
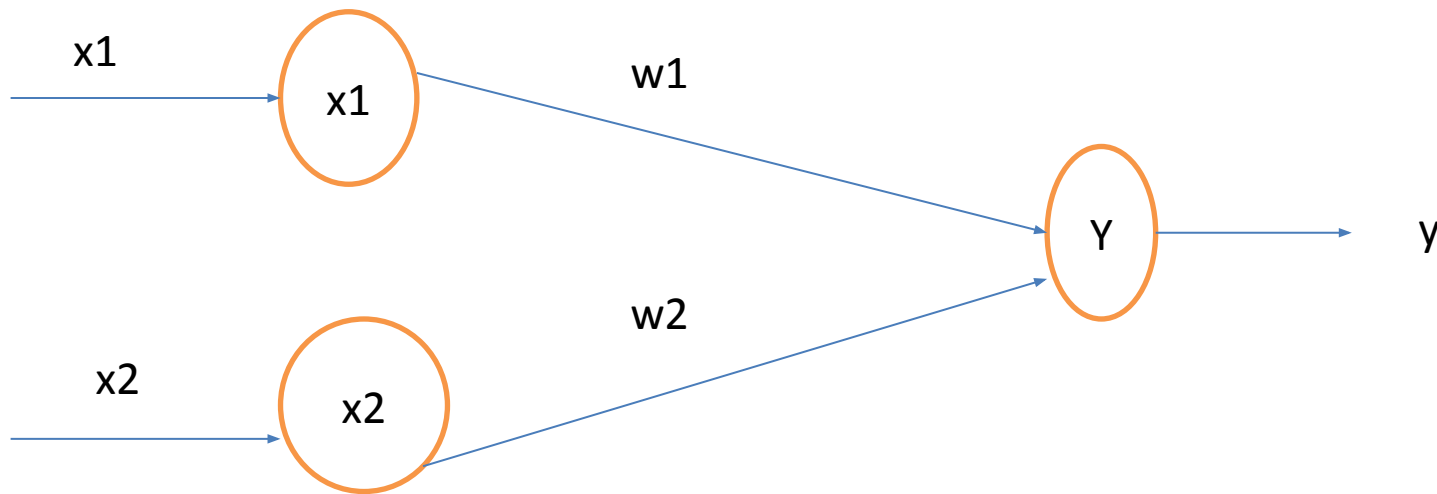


Neural Network and Natural Language Processing

Artificial neural network

- Basic components of artificial neural network are:
 - Neurons (Nodes)
 - Layers
 - Connections
 - Activation function
- In neurons some values or data is stored.





$$Y = w1 * x1 + w2 * x2$$

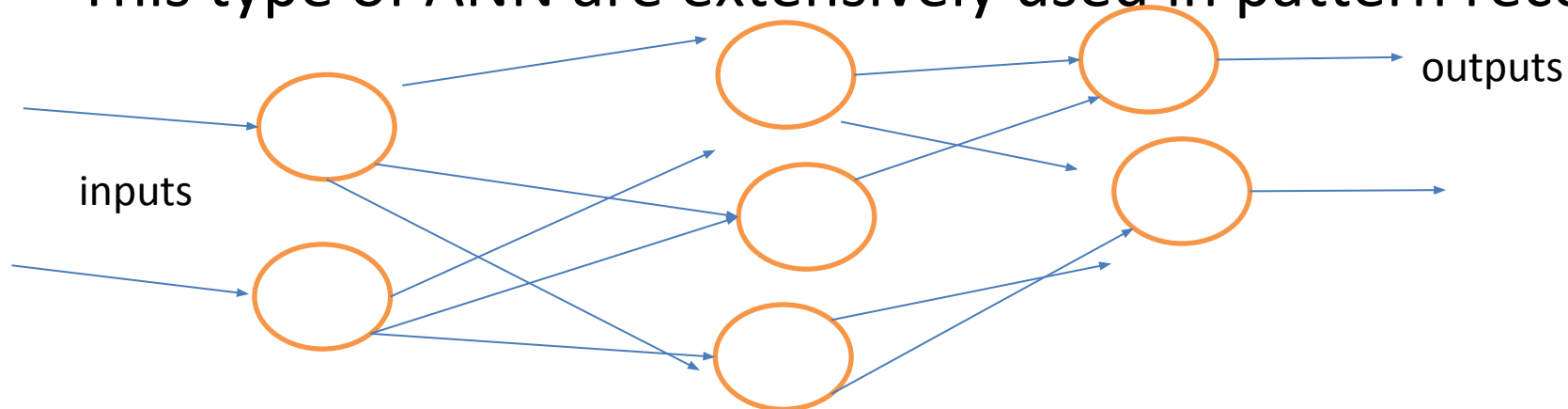
Sigmoid function is used
as activation function
So $\sigma(y) = 1 / (1 + e^{-y})$

- For eg
 - $w_1 = 0.5$
 - $w_2 = -0.3$
 - $b = 0.1$
- This value is taken after number of training in neural network.

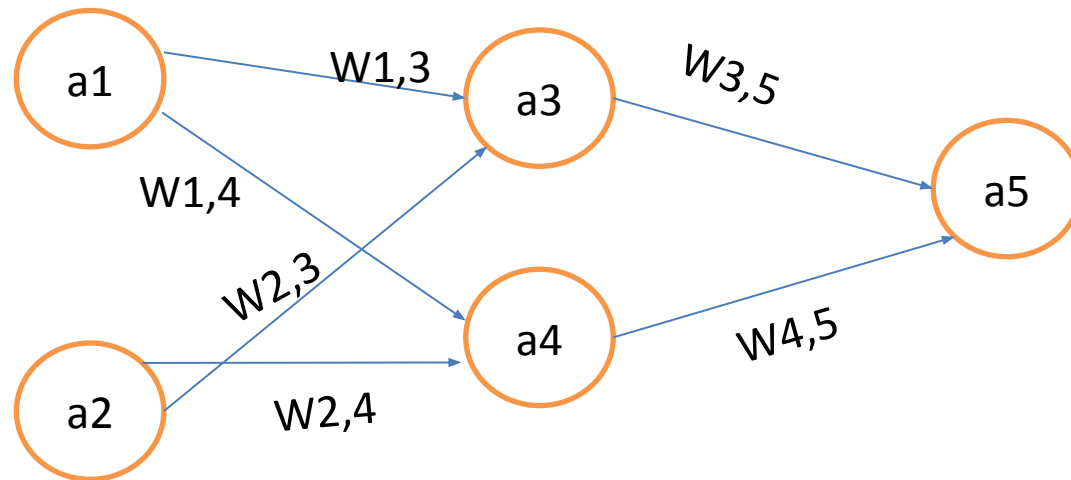
- Activation function: the process of deciding output after aggregating input is called activation function.
- Types:
- Linear function: for linear activation function, the output activity is proportional to the total weighted output.
 - $g(x)=k*x+c$, where k and c are constant.
- Threshold function: for threshold function, the output are set at one of the two levels, depending on whether the total input is greater than or less than some threshold value.
 - $g(x)=1$, if $x \geq k$
 - $g(x)=0$, if $x < k$
- Sigmoid function: For sigmoid activation function, the output varies continuously but not linearly as the input changes.
 - $\sigma(y) = 1/(1 + e^{-y})$

Types of ANN

- Feed-forward network:
 - feed-forward ANN allows signal to travel one way only: from input to output.
 - There is no feedback(loops) i.e the output of any layer doesnot affect the same layer.
 - This type of ANN are extensively used in pattern recognition.



Example of feedforward



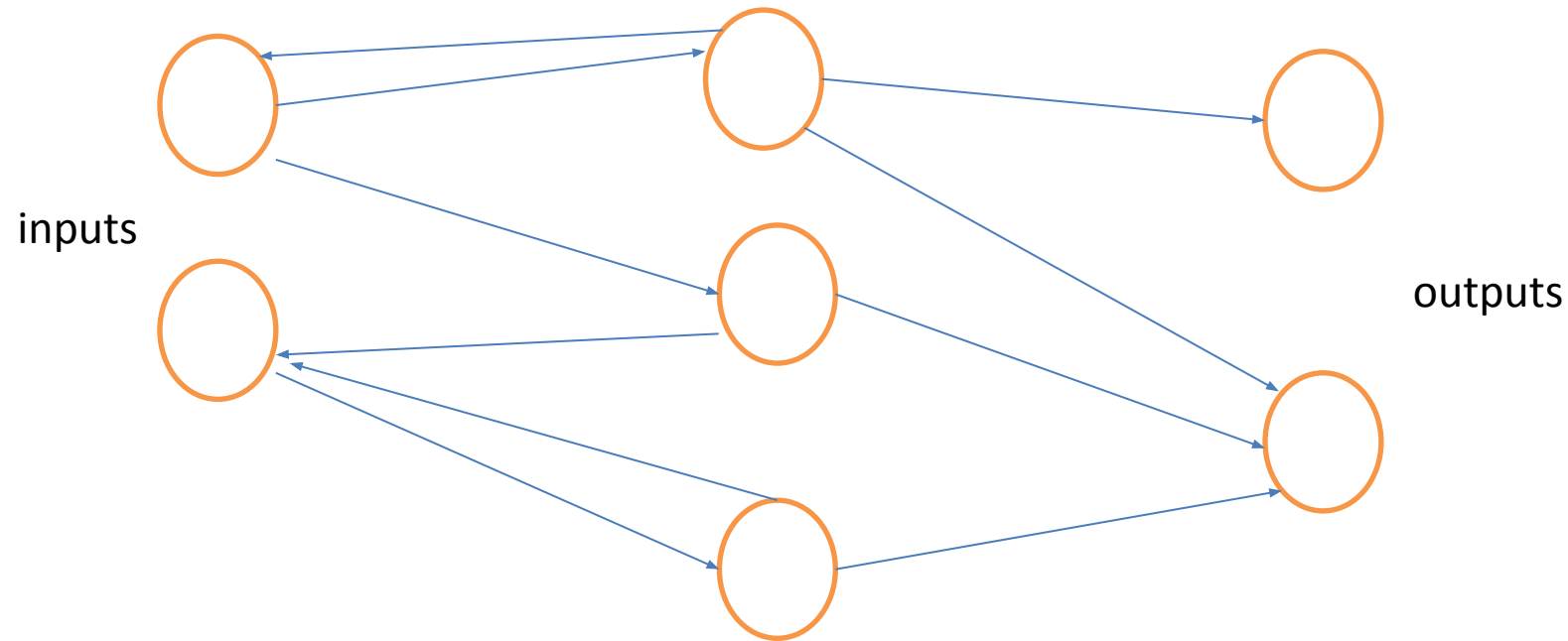
$$g_3 = g(w_{1,3} * a_1 + w_{2,3} * a_2)$$

$$g_4 = g(w_{1,4} * a_1 + w_{2,4} * a_2)$$

$$g_5 = g(w_{3,5} * a_3 + w_{4,5} * a_4)$$

Feedback networks (recurrent networks)

- Feedback networks can have signal travelling in both direction by introducing loops in the network.



Application of ANN

- Image and Video Processing:
 - Applications:
 - Face recognition (e.g., Facebook photo tagging)
 - Object detection (e.g., autonomous vehicles)
 - Medical imaging (e.g., tumor detection in X-rays or MRIs)
- Natural Language Processing (NLP):
 - Applications:
 - Machine translation (e.g., Google Translate)
 - Text classification (e.g., spam detection)
 - Sentiment analysis (e.g., product reviews)
- Speech Recognition and Audio Processing:
 - Applications:
 - Voice assistants (e.g., Google Assistant, Alexa)
 - Automatic transcription (e.g., YouTube captions)

- Healthcare and Bioinformatics:
 - Applications:
 - Disease prediction (e.g., cancer, diabetes)
 - Drug discovery
- Finance and Banking
 - Applications:
 - Stock market prediction
 - Credit scoring
 - Fraud detection

Boltzmann Machine (BM)

- A Boltzmann Machine is a type of stochastic recurrent neural network that uses a probabilistic approach to model data distributions i.e Instead of memorizing the training data, the Boltzmann Machine learns how likely different patterns are.
- It is mainly used for unsupervised learning, meaning it learns patterns from input data without requiring labeled outputs.
- It consists of a set of neurons connected in a network, where each neuron can take binary states (0 or 1).
- These networks are energy-based models that learn by minimizing an energy function, similar to how physical systems reach equilibrium.

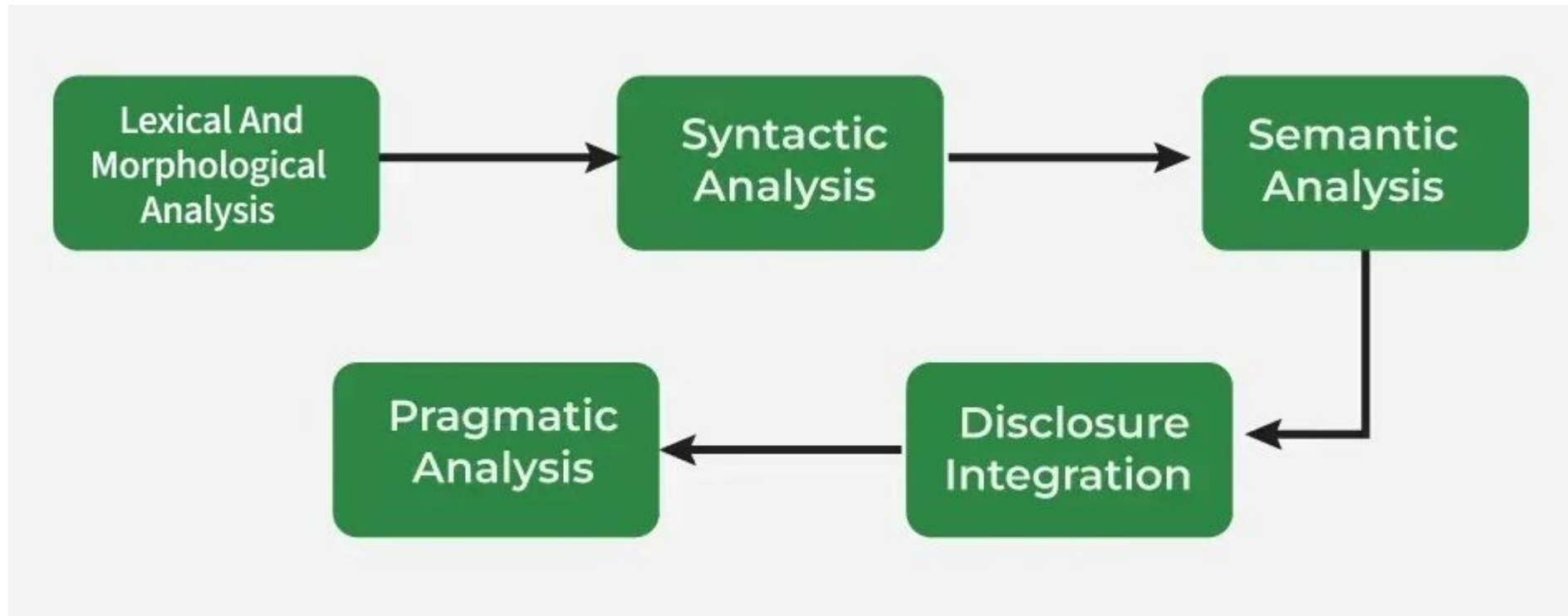
- Low energy → Stable and more likely.
- High energy → Unstable and less likely.

Concepts of natural language understanding and natural language generation

- **Natural Language Processing (NLP)** is divided into two major areas:
 - **Natural Language Understanding (NLU)** – Understanding human language.
 - **Natural Language Generation (NLG)** – Generating human language.
- Natural Language Understanding (NLU) is the branch of AI that enables a computer to **understand the meaning of human language** (text or speech).
- It converts **human language → machine-understandable representation**.
- **Example**
- Input:
- "Book me a flight from Kathmandu to Pokhara tomorrow."
- The computer should understand:
- Intent: Book flight
- Source: Kathmandu
- Destination: Pokhara
- Date: Tomorrow
- The computer does **not** simply see words—it understands what they mean.

- Natural Language Generation (NLG) is the branch of AI that **converts machine data or internal representations into human-readable language.**
- It converts **machine representation → human language.**
- **Example**
- Computer data:
- Temperature = 35°C Weather = Sunny
- Generated sentence:
- "Today's weather is sunny with a temperature of 35°C."

Steps in Natural Language Processing



Lexical Analysis (Word-Level Analysis)

- Lexical analysis is the first step of NLP. It breaks the input sentence into words or tokens and identifies their basic properties.
- Example
- Sentence:
- "The students are studying."
- After lexical analysis:

Word	Part of Speech
The	Article
students	Noun
are	Verb
studying	Verb
Output: A list of meaningful words with their grammatical categories.	

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The	Article
students	Noun
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Output: A list of meaningful words with their grammatical categories.	

Morphological Analysis

- Morphological analysis studies the internal structure of words using morphemes, the smallest units of meaning.
- Stemming reduces words to their root form like "running" to "run".
- Lemmatization converts words to their base or dictionary form considering the context like "better" becomes "good".

Syntactic Analysis (Parsing)

- Syntactic analysis checks whether the sentence follows the grammatical rules of the language and determines the relationship between words.
- **Example**
- Sentence:
- "Ram eats an apple."
- Structure:
- Subject : Ram
- Verb : eats
- Object : apple
- If the sentence is:
- "Apple Ram eats"
- The parser recognizes that the word order is grammatically incorrect in English.
- **Output:** The grammatical structure of the sentence.

Semantic Analysis (Meaning Analysis)

- Semantic analysis determines the **meaning** of the sentence after its grammatical structure has been identified.
- **Example**
- Sentence:
- "Ram drinks water."
- Computer understands:
- Person : Ram, Action : Drink, Object : Water
- Another example:
- "The bank is closed."
- Semantic analysis tries to determine whether **bank** means:
- A financial institution, or
- The side of a river.
- **Output:** The meaning of the sentence.

Discourse Integration

- Discourse integration connects the current sentence with previous sentences to understand the overall context.
- **Example**
- Ram bought a laptop. He uses it every day.
- The computer identifies:
- **He** → Ram
- **it** → Laptop
- Without discourse analysis, the pronouns would be ambiguous.
- **Output:** Context-aware understanding across multiple sentences.

Pragmatic Analysis

- Pragmatic analysis determines the speaker's **intended meaning** using context, background knowledge, and real-world situations.
- Teacher says:
- "Can you open the window?"
- Literal meaning:
 - Asking whether the student has the ability.
 - Actual meaning:
 - A polite request to open the window.